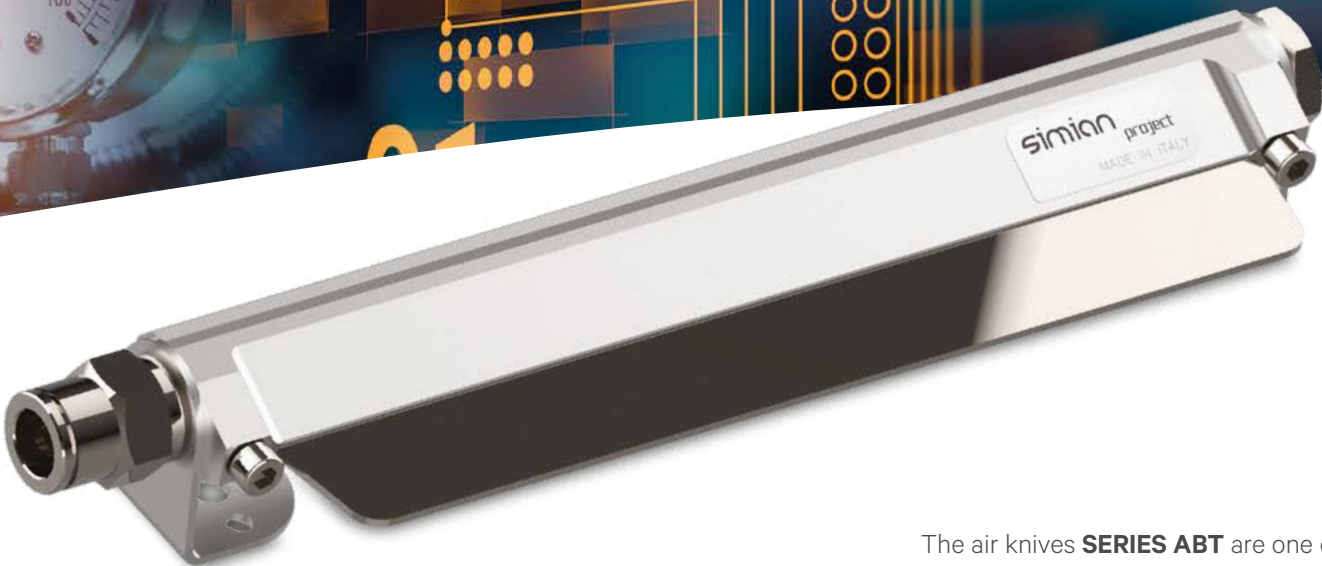
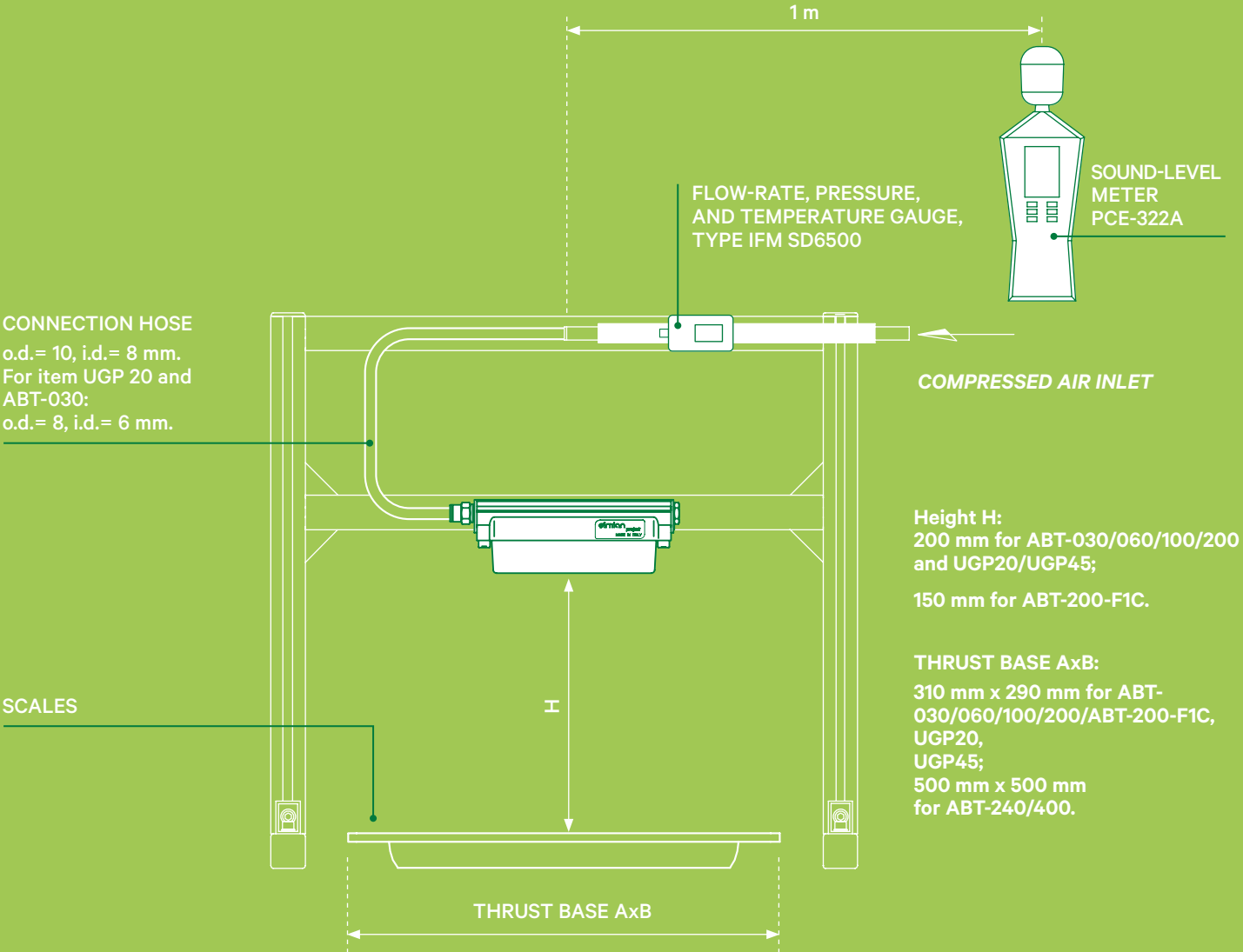




Description of the set-up and instrumentation used for our tests on air knives.



The air knives **SERIES ABT** are one of a kind, thanks to their high blowing power, which is a result of the air flow on both sides of the blade, and thanks to their easy installation, by means of two neodymium magnets and of brackets, which make it possible to direct the knife according to all demands. These products are very effective for cleaning, drying, and cooling.

- Design geometries optimised to maximise the Coanda effect
- Double blow-off flow (both sides of the blade)
- Powerful, uniform flow, suitable for cleaning small and large surfaces
- Modular design and possibility of customisation
- No moving parts, so maintenance-free

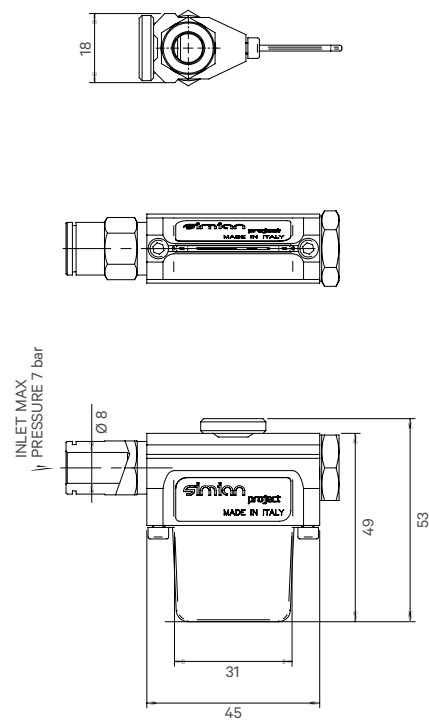
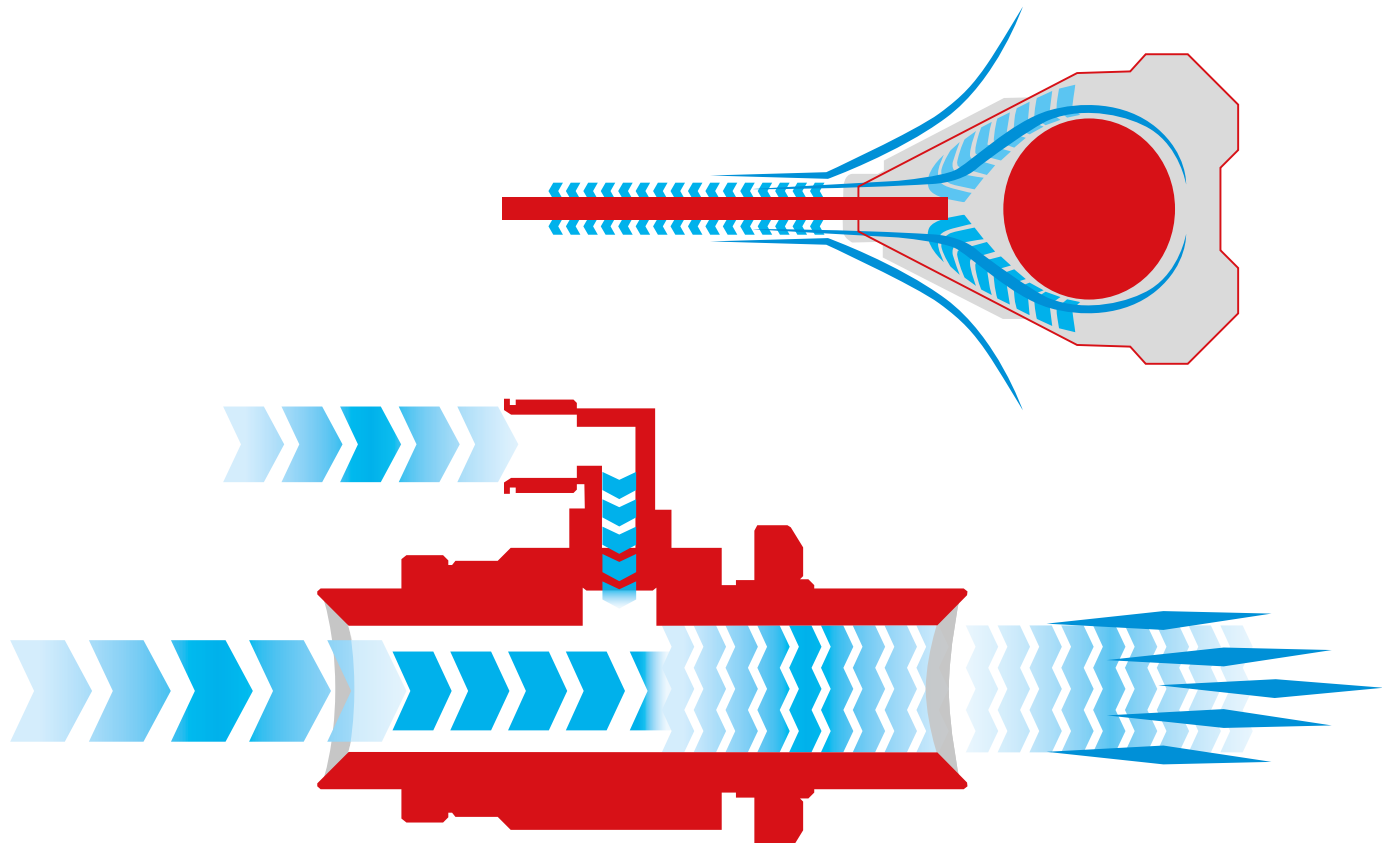


DESCRIPTION OF THE COANDA EFFECT

The air amplifiers and the air knives exploit the Coanda effect.

This phenomenon can be explained as the tendency of a fluid to follow the contour of a surface nearby. It is named after the pioneer of aerodynamics Henri Coanda, who in 1936 patented some instruments that exploited the capacity to deviate a flow.

The compressed air introduced in an amplifier or in an air knife is forced to pass through a reduced section, from 0.02 mm to 0.08 mm, and, by lapping the surface nearby, the surrounding air is attracted towards the flow's direction, so that the volume of air becomes from 5 to 20 times bigger than it was at the inlet.



GENERAL FEATURES - ABT-030

Materials	Anodized aluminium and AISI304 s.s.
Air supply port	Fitting Ø-8
Fixation	Optional angular bracket
Blade length	32 mm
Air supply pressure	1-7 bar
Optional magnetic support	KACM-ABT030
Weight	110 g

PERFORMANCES AND CONSUMPTION TABLE

Pressure bar	Consumption NI/min	Thrust a 200 mm in g	Noise level dBA
1	150	97	70
2	255	213	76
3	346	330	79
4	433	450	82
5	516	590	84
6	599	720	85
7	666	850	86

SERIES ABT-030 PLUS

DOUBLE-SLOT AIR KNIVES



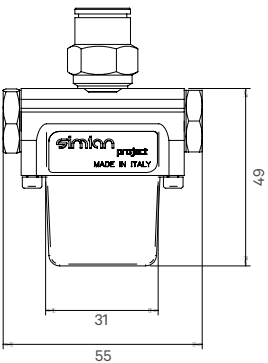
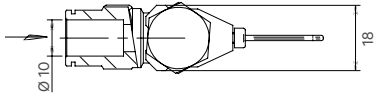
SERIES ABT-060

DOUBLE-SLOT AIR KNIVES



GENERAL FEATURES - ABT-030 PLUS

Materials	Anodized aluminium and AISI304 s.s.
Air supply port	Fitting Ø-10
Fixation	Optional angular bracket
Blade length	32 mm
Air supply pressure	1-7 bar
Optional magnetic support	KACM-ABT030
Weight	130 g

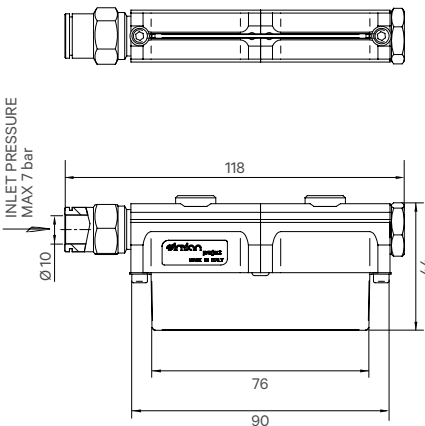
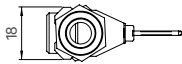


PERFORMANCES AND CONSUMPTION TABLE

Pressure bar	Consumption NI/min	Thrust a 200 mm in g	Noise level dBA
1	166	120	70
2	266	230	76
3	366	360	78
4	458	500	81
5	549	640	82,5
6	633	780	84
7	716	940	86

GENERAL FEATURES - ABT-060

Materials	Anodized aluminium and AISI304 s.s.
Air supply port	Fitting Ø-10
Fixation	Optional angular bracket
Blade length	76 mm
Air supply pressure	1-7 bar
Optional magnetic support	KACM-ABT030
Weight	170 g



PERFORMANCES AND CONSUMPTION TABLE

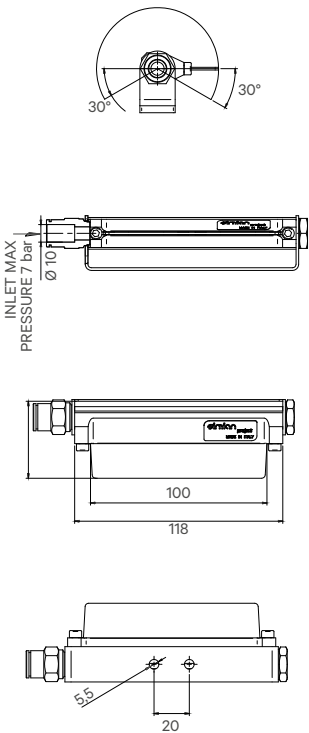
Pressure bar	Consumption NI/min	Thrust a 200 mm in g	Noise level dBA
1	283	170	72
2	449	340	77
3	629	570	80
4	816	810	83
5	982	1090	85
6	1166	1400	86
7	1350	1700	87

SERIES ABT-100

DOUBLE-SLOT AIR KNIVES



GENERAL FEATURES - ABT-100	
Materials	Anodized aluminium and AISI304 s.s.
Air supply port	Fitting Ø-10
Fixation	Integrated feet
Blade length	100 mm
Air supply pressure	1-7 bar
Optional magnetic support	KACM-ABT100
Weight	200 g



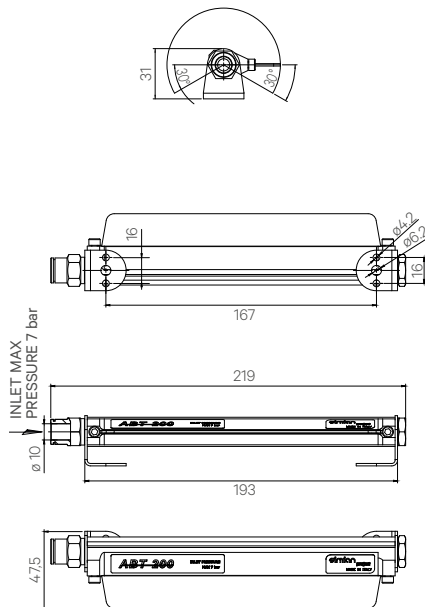
PERFORMANCES AND CONSUMPTION TABLE				
Pressure bar	Consumption NI/min	Thrust a 200 mm in g	Noise level dBA	
1	366	216	75	
2	558	450	79	
3	724	690	82	
4	899	930	84	
5	1082	1210	85	
6	1233	1530	87	
7	1410	1800	88	

SERIES ABT-200

DOUBLE-SLOT AIR KNIVES



GENERAL FEATURES - ABT-200	
Materials	Anodized aluminium and AISI304 s.s.
Air supply port	Fitting Ø-10
Fixation	Integrated feet
Blade length	170 mm
Air supply pressure	1-7 bar
Optional magnetic support	KACM-ABT200
Weight	290 g



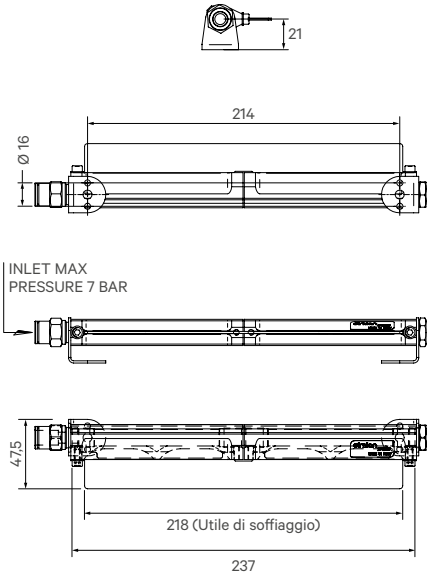
PERFORMANCES AND CONSUMPTION TABLE				
Pressure bar	Consumption NI/min	Thrust a 200 mm in g	Noise level dBA	
1	499	245	75	
2	940	570	80	
3	1582	1030	84	
4	2082	1450	87	
5	2665	2000	90	
6	3248	2400	93	

SERIES ABT-240

DOUBLE-SLOT AIR KNIVES



GENERAL FEATURES - ABT-240	
Materials	Anodized aluminium and AISI304 s.s.
Air supply port	Fitting Ø 10
Fixation	Integrated feet
Flow width	218 mm
Supply pressure	1-7 bar
Optional magnetic support	KACM-ABT100
Weight	300 g



PERFORMANCES AND CONSUMPTION TABLE

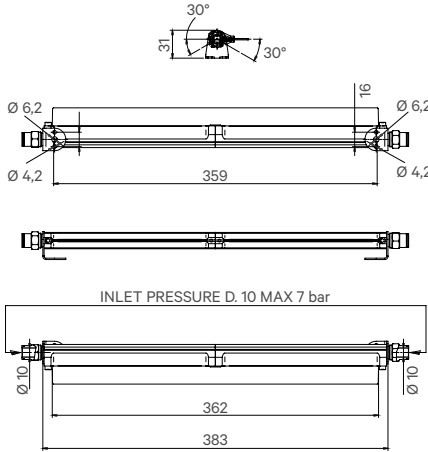
Pressure bar	Consumption NI/min	Thrust a 200 mm in g	Noise level dBA
1	283	190	72
2	449	350	77
3	629	515	80
4	816	730	83
5	982	990	85
6	1166	1260	86

SERIES ABT-400

DOUBLE-SLOT AIR KNIVES



GENERAL FEATURES - ABT-400	
Materials	Anodized aluminium and AISI304 s.s.
Air supply port	Fitting Ø 10
Fixation	Integrated feet
Flow width	362 mm
Supply pressure	7 bar
Optional magnetic support	KACM-ABT200
Weight	450 g



PERFORMANCES AND CONSUMPTION TABLE

Pressure bar	Consumption NI/min
1	1530
2	1956
3	2720
4	3146
5	3910
6	4760
7	5326

SERIES ABT-600

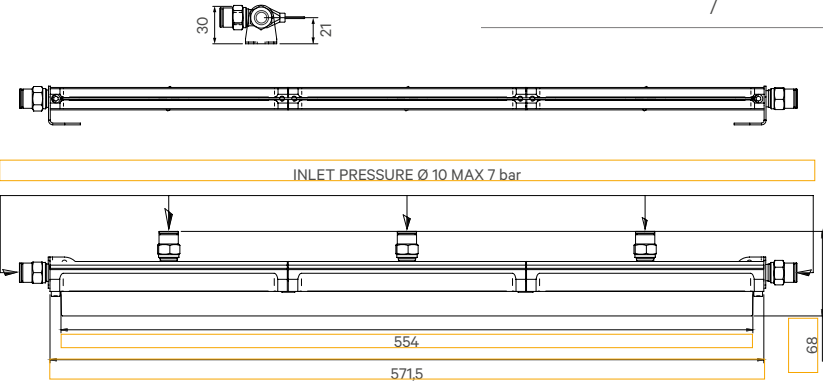
DOUBLE-SLOT AIR KNIVES



GENERAL FEATURES - ABT-600	
Materials	Anodized aluminium and AISI304 s.s.
Air supply port	Fitting Ø 10
Fixation	Integrated feet
Flow width	554 mm
Supply pressure	max 7 bar
Optional magnetic support	KACM-ABT200
Weight	690 g

PERFORMANCES AND CONSUMPTION TABLE

Pressure bar	Consumption NI/min
1	2295
2	2934
3	4080
4	4719
5	5865
6	7140
7	7989



SERIES ABT-800

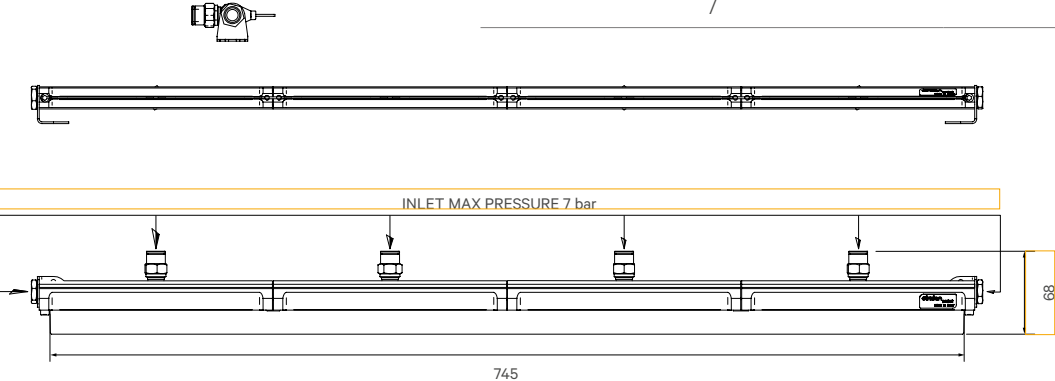
DOUBLE-SLOT AIR KNIVES



GENERAL FEATURES - ABT-800	
Materials	Anodized aluminium and AISI304 s.s.
Air supply port	Fitting Ø 10
Fixation	Integrated feet
Flow width	745 mm
Supply pressure	max 7 bar
Optional magnetic support	KACM-ABT200
Weight	900 g

PERFORMANCES AND CONSUMPTION TABLE

Pressure bar	Consumption NI/min
1	2295
2	2934
3	4080
4	4719
5	5865
6	7140
7	7989



ACCESSORIES AND SPECIAL VERSIONS

AIR KNIVES

SERIES ABT-F1C

CALIBRATED SINGLE-SLOT AIR KNIVES



ABT 200
WITH CONCAVE AND CONVEX LIP



ABT 030 PLUS
WITH SHAPED LIP



ABT 030 PLUS
WITH TUBE FOR CONNECTION
TO MACHINE TOOL



ABT 030 PLUS
WITH CURVED LIP



GENERAL FEATURES - ABT-F1C

Materials	Aluminium and Delrin
Air supply port	Fitting Ø 10
Fixation	Integrated feet
Flow width	From 150 mm to 1200 mm
Supply pressure	max 7 bar
Noise level for every 200 mm length	Range from 62 to 82 dBA
Optional magnetic kit	KACM-ABT200

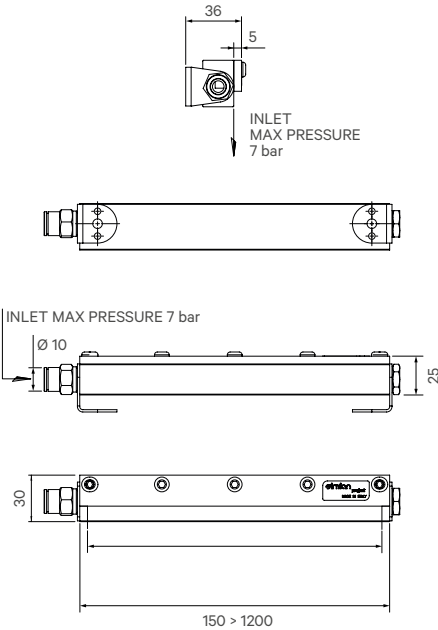
PERFORMANCE AND CONSUMPTION TABLE (For length 100 mm)

Pressure bar	Consumption NI/min	Thrust force at 150 mm, in g
1	109	61
2	195	135
3	279	220
4	341	310
5	416	400
6	483	520
7	566	635

MAGNETIC SUPPORT	
PART-NUMBER	AIR KNIFE
KACM-ABT030	ABT-030 / ABT-060
KACM-ABT100	ABT-100
KACM-ABT200	ABT-200 / ABT-240 / ABT-400 / ABT-600 / ABT-800

FIXATION BRACKET	
PART-NUMBER	AIR KNIFE
ABT-05	ABT-030 / ABT-060

CUSTOMISED VERSIONS	
On request we can supply versions customised in shape and/or material.	

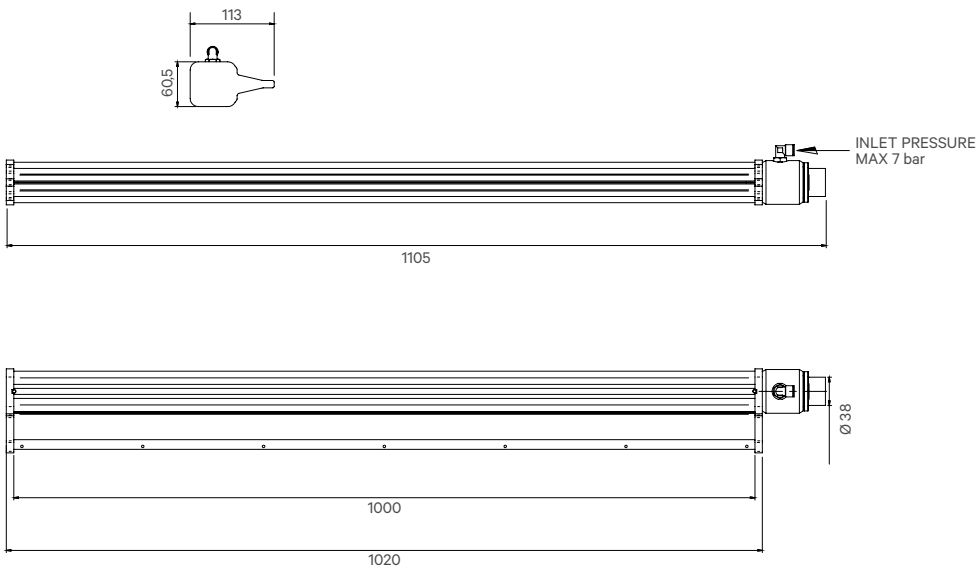


SERIES ABX-1000

LONG AIR KNIVES SUPPLIED BY AIR AMPLIFIER



GENERAL FEATURES - ABX-1000	
Materials	Anodised aluminium / Stainless steel
Fixation	On request
Flow width	1000 mm
Supply pressure	Based on the type of air supply
On request: supply connection at 90°.	

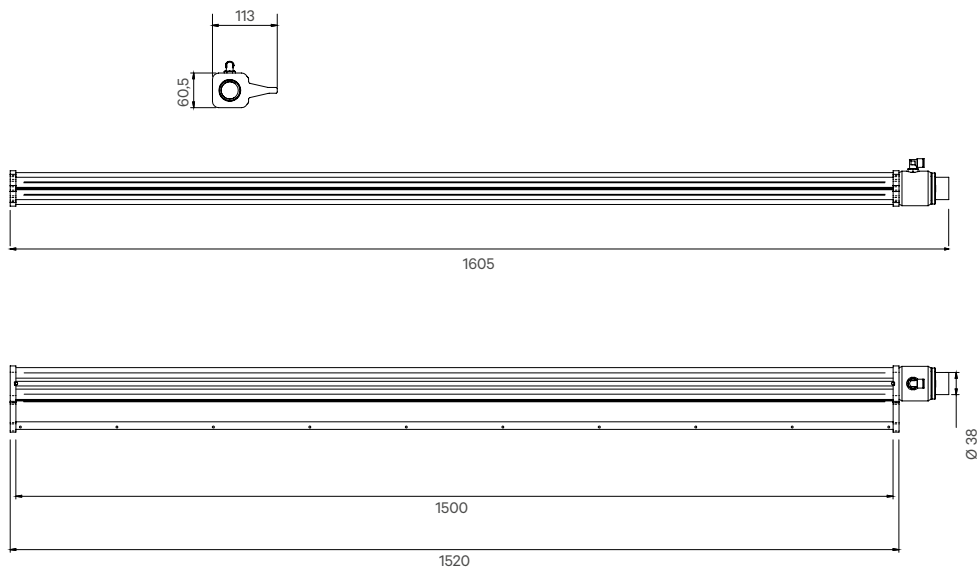


SERIES ABX-1500

LONG AIR KNIVES SUPPLIED BY AIR AMPLIFIER



GENERAL FEATURES - ABX-1500	
Materials	Anodised aluminium / Stainless steel
Fixation	On request
Flow width	1500 mm
Supply pressure	Based on the type of air supply
On request: supply connection at 90°.	



SERIES ABX-2000

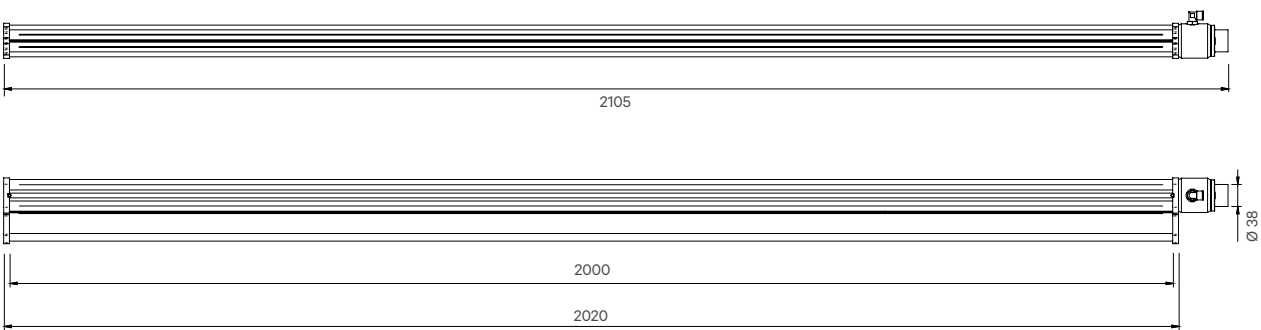
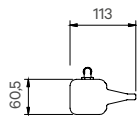
LONG AIR KNIVES SUPPLIED BY AIR AMPLIFIER



GENERAL FEATURES - ABX-2000

Materials	Anodised aluminium / Stainless steel
Fixation	On request
Flow width	2000 mm
Supply pressure	Based on the type of air supply

On request: supply connection at 90°.



SERIES ABZ-1000

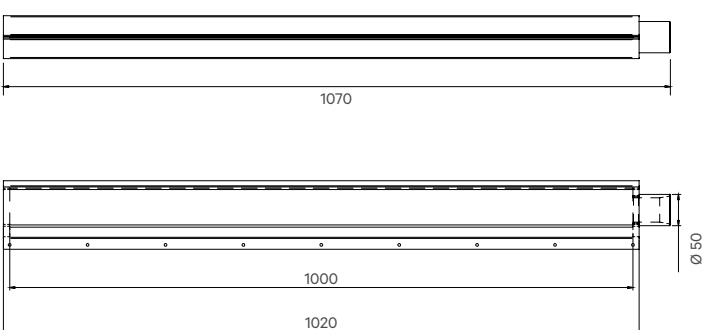
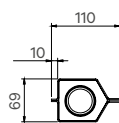
LONG AIR KNIVES TO BE SUPPLIED BY ELECTRIC BLOWER



GENERAL FEATURES - ABZ-1000

Materials	Zinc-plated metal sheet
Fixation	On request
Flow width	1000 mm
Supply pressure	Based on the type of air supply

On request: supply connection at 90°.



SERIES ABZ-1500

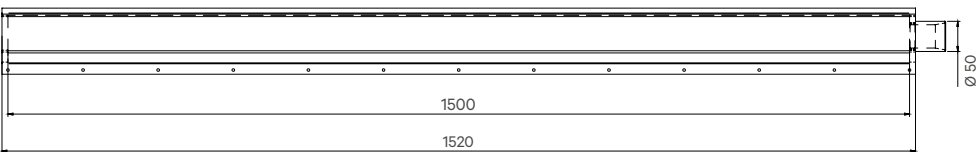
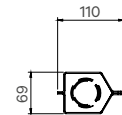
LONG AIR KNIVES TO BE SUPPLIED BY ELECTRIC BLOWER



GENERAL FEATURES - ABZ-1500

Materials	Zinc-plated metal sheet
Fixation	On request
Flow width	1500 mm
Supply pressure	Based on the type of air supply

On request: heated air flow.



SERIES ABZ-2000

LONG AIR KNIVES TO BE SUPPLIED BY ELECTRIC BLOWER



GENERAL FEATURES - ABZ-2000

Materials	Zinc-plated metal sheet
Fixation	On request
Flow width	2000 mm
Supply pressure	Based on the type of air supply

On request: heated air flow.

